

Abundant stocks and mobilization of elements in boreal acid sulfate soils



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ABSTRACT

Large amounts of sulfate and divalent iron (Fe) are released into the pore water of acid sulfate (AS) soils upon oxidation of sulfidic materials. The simultaneously produced acidity dissolves metals from the soil matrix. Reduced horizons of AS soils commonly have a large mineral nitrogen (N) stock in the form of NH_4^+ -N, which is a potential source of N leaching and gaseous emissions. This study was carried out at three AS soil sites in Finland. Cation composition of pore water was monitored in a monolithic lysimeter experiment. Timothy samples grown in an AS soil were analysed for mineral elements and the soil was investigated for zinc (Zn) distribution in different chemical species at four depths down to 85 cm. The composition of pipestems formed in previous root channels was investigated by SEM, X-ray EDX and XRD. Emissions of CO_2 and N_2O were measured in an AS field which had a peaty topsoil. Monitoring by closed chambers was carried out at three sites differing in the depth of peat layer (15, 30 and 60 cm). In the sulfuric horizon, large amounts of calcium and magnesium were mobilized. Reflooding stopped the mobilization of those elements but resulted in abundant dissolution of Fe, which became the dominant cation in the pore water. Timothy growing in an AS soil showed only small deviations from the average composition measured in Finland with the exception of Zn that was at the deficiency level. As Zn in the root zone had been dissolved and subsequently leached, it was recovered in easily soluble forms in the subsoil. Pipestem composition indicated accumulation of Fe and formation of a new solid phase probably mostly in the form of schwertmannite even though jarosite was also detected. The annual CO_2 emissions from the peaty AS soil were about $6000 \text{ kg C ha}^{-1}$ but the N_2O emissions were relatively small, with a magnitude more typical of mineral rather than organic soils. Thus, large N stock of an AS soil do not necessarily contribute to abundant gaseous N emissions.

1. Introduction

Substantial changes in soil chemistry take place upon natural or artificial drainage and subsequent oxidation of sulfidic materials in acid sulfate (AS) soils. Sulfate and divalent Fe are released into the pore water from the oxidizing FeS/FeS_2 . The pH is lowered from neutrality to values below 3.5, dissolving heavy metals and Al from the solid matrix. Part of the metals (at least Ni, Mn, Co and Cu) also originates from the oxidizing FeS_2 (Sohlenius and Öborn, 2004; Claff et al., 2011). Owing to low pH, many metals released from the solid phase remain

dissolved in pore water in elevated concentrations resulting in increased leaching of these substances. Abundance of dissolved Al^{3+} displaces base cations from exchange sites and subjects them to leaching (Hartikainen and Yli-Halla, 1986). Gradually the oxidized horizons become saturated with exchangeable Al (Dent, 1986; Yli-Halla, 1997; Yvanes-Giuliani et al., 2014). Leaching of Al itself occurs in abundant amounts only after most exchange sites are occupied with Al^{3+} (Hartikainen and Yli-Halla, 1986; Shamshuddin et al., 2004; Virtanen, 2015). The stock of Al in soil is inexhaustible. There is a close relationship between acidity and sulfate concentration as well as Al and

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Fig. 1. The locations of the experimental sites. The shading around the Baltic Sea stands for the area that was under the Litorina Sea and is thus the potential area for the coastal acid sulfate soils.

sulfate concentration in stream and river water (Toivonen and Österholm, 2011), and strong negative relationships between pH and Al concentration in pore water, discharge water and groundwater in AS soils have been found (Virtanen et al., 2014b). Österholm and Åström (2002) estimated that within 2 m of soil surface the stock of S that can be leached from a typical AS soil of Rintala area in Finland was about 40 tons ha^{-1} . In this particular AS soil area about 630 kg S ha^{-1} was annually leached (Österholm and Åström, 2004) while in non-AS arable soils leaching has amounted to 15 kg ha^{-1} (Turtola and Jaakkola, 1986) or 20 kg ha^{-1} (Eriksen and Askegaard, 2000) in Finland and Denmark, respectively. Concentrations of metals in Finnish streams draining AS soils are much higher than in other streams (e.g., Åström and Björklund, 1995; Åström and Corin, 2000; Toivonen and Österholm, 2011). The total metal concentration in sulfidic sediments of Sweden and Finland is not higher than in other soils but the metals seem to be mobilized to a larger extent (Sohlenius and Öborn, 2004). Therefore, it has been hypothesized that plants growing in these soils may be high in heavy metals, but little proof has thus far been found (Palko, 1986; Fäلتmarsch et al., 2010).

Translocation of Fe out of the soil profile is not as marked as that of many other metals (Sohlenius and Öborn, 2004) and Fe concentrations of streams draining AS soils do not differ substantially from streams in non-AS soils (Åström and Corin, 2000; Sundström et al., 2002). Instead, upon oxidation of Fe^{2+} to Fe^{3+} new solid phases are precipitated as continuous coatings or pipestems around former root channels, with Fe released from FeS/FeS_2 being redistributed within the soil profile. However, in reflooded soils which contain plenty of precipitated Fe oxides, concentration of Fe^{2+} in the pore water can increase substantially (e.g. Johnston et al., 2014; Virtanen et al., 2014a) and result in formation of Fe sulfides (e.g. Burton et al., 2011; Johnston et al.,

2011) or leaching of Fe into watercourses (Cook et al., 2000; Virtanen, 2015). At the initial stages of oxidation and leaching, intermediate oxidation products such as jarosite $\text{KFe}_3(\text{OH})_6(\text{SO}_4)_2$ and schwertmannite $\text{Fe}_8\text{O}_8(\text{OH})_6(\text{SO}_4)_n\text{H}_2\text{O}$ are commonly identified in AS soils, and upon further leaching, they are gradually transformed into Fe hydroxides. The pipestems have a marked effect on the hydraulic properties of an AS soil. While the reduced subsoil is practically impermeable, the pipestems and permanent desiccation cracks stabilized by Fe hydroxides create effective transport routes for substances in the oxidized horizons (Dent, 1986; Joukainen and Yli-Halla, 2003; Johnston et al., 2009).

Abundant reserves of NH_4^+ -N have been measured in the reduced subsoil, creating potential for N leaching or gaseous N losses. In western Finland Paasonen-Kivekäs and Yli-Halla (2005) measured as much as 435 kg ha^{-1} mineral N in the upper 240 cm of an AS soil while 137 kg ha^{-1} was measured in an adjacent non-AS soil. Results of similar magnitude have been obtained also in an AS soil in Helsinki, Finland by Šimek et al. (2011), who suggested that NH_4^+ -N originates from slow decomposition of organic matter and accumulates in the subsoil because nitrification does not take place in those permanently water-logged conditions. Uusi-Kämpä et al. (2012) measured N losses of as much as 50 kg ha^{-1} in drainage water from an AS soil while the average amount leached from soils of Finland is about 16 kg ha^{-1} (Tattari et al., 2017). The abundance of plant available N in soil was also demonstrated by the fact that wheat yields contained 140–150 kg N ha^{-1} while only 90 kg N ha^{-1} had been applied in fertilizer (Uusi-Kämpä et al., 2012).

Often stocks, fluxes and transformations of different substances are investigated one by one in separate studies. In this paper, we try to pull together information of multiple elements from boreal acid sulfate soils

in Finland. These data mostly reflect abundant fluxes of many substances in AS soils induced by oxidation and acidification. In terms of Fe, the abundant flux is caused by a response to reflooding and return of reduced conditions.

2. Materials and methods

2.1. Experimental sites, experiments and samples

The results in this paper come from three sites, Ylistaro, Helsinki and Joensuu in W, S and E Finland, respectively (Fig. 1). Sulfic Cryaquepts of Ylistaro (Yli-Halla, 1997) and Helsinki (Šimek et al., 2011; Virtanen et al., 2013) are AS soils sedimented during the Litorina Sea stage (c. 7850–2750 years BP; Andren et al., 2011) of the Baltic Sea. The Typic Sulfaquept/Typic Sulfosaprist in Joensuu is a former peat mining area of Pärnäsuo where the mineral subsoil originating from black schists (Loukola-Ruskeeniemi et al., 1991) has been exposed to rapid oxidation after peat removal (Mäkelä and Yli-Halla, 2014). The mean annual air temperature in Helsinki is 6 °C, in Ylistaro 4 °C and in Joensuu 3 °C. The precipitation is about 650 mm, 550 mm and 600 mm respectively, about half being evaporated and half drained away.

The Helsinki and Ylistaro soils have been comprehensively described by Virtanen et al. (2013) and Yli-Halla (1997), respectively, and properties relevant in this paper are shown in Table 1. The Helsinki soil (60° 13' N, 25° 00' E, at sea level) by the sea was poldered for agricultural use in the 1950's and was artificially drained in 1958 with subsurface pipes. The soil has a strong structure stabilized by iron hydroxide coatings down to about 125 cm, which is the starting depth of the massive soil containing sulfidic materials. The Helsinki soil is an example of a very young AS soil. The Ylistaro soil (62° 55' N, 22° 29' E, elevation 26 m) is about 50 km from the current shore line and represents the geographical area where the AS soils are the most prevalent in Finland. The soil surface emerged from the sea about 2500 years ago owing to the on-going post-glacial land uplift and has a cultivation history of > 100 yrs. The Ylistaro soil has a strong structure down to about 2 m and massive soil with sulfidic materials occurs below 210 cm. This soil is thus in a more advanced state of development compared to the Helsinki soil. In the Joensuu site (62° 25' N, 30° 04' E, elevation 83 m), peat was excavated for 30 years, and since 2011 the area has been in agricultural use and drained with open ditches. When peat excavation ended, the thickness of peat varied from non-

existent to 0.7 m. Compared to the other sites, the Joensuu soil has a rather weak structure. Jarosite occurs in old root channels and traces of sulfidic materials occur below 50 cm. Among the three experimental sites, the Joensuu soil represents a different type of AS soils because the sulfidic materials originate from the ancient black schists ground from the rock by the continental ice during the Weichselian glaciation.

The response of pore water composition to the gradual oxidation of sulfidic material was monitored in the Helsinki soil over 28 months in 1-m deep monolithic lysimeters (Virtanen et al., 2013), which were sealed from the bottom to prevent artefact oxidation of sulfidic materials. The monoliths consisted of Ap, Bg, Bgic, BCg and Cg horizons. Two water management treatments were studied. In the high ground-water treatment (HGW, four replicates) water level was at 20–30 cm from soil surface throughout the year, thus reflooding the oxidized and acidified subsoil horizons. The low groundwater treatment (LGW, three replicates) mimicked natural groundwater variation in fields. During summer, water table was allowed to drop down to 70 cm, i.e., into the BCg horizon, due to drainage and evapotranspiration but in autumn the water table was elevated to 40–50 cm (Bg horizon) according to the natural conditions in fields. Reed canary grass (*Phalaris arundinacea*) was growing in the lysimeters. Moreover, there were two HGW lysimeters without plants but those results are not presented in this paper. Pore waters from each horizon were extracted by fixed Rhizon samplers (MOM, Ø 0.15 µm in porous material) into 10-ml plastic vacuum tubes daily during intensive measurement campaigns and otherwise every second week in summer and once a month in winter. Elements were analysed by ICP-OES (Thermo Scientific, ICAP 6000) explained in detail in Virtanen et al. (2014a) and Virtanen (2015). The mean element concentrations of pore water and the standard deviations were calculated for the following periods: 22.8–30.9.2008, 1.10.2008–31.1.2009, 1.2.–31.5.2009, 1.6.–30.9.2009, 1.10.2009–31.1.2010, 1.2.–31.5.2010, 1.6.–30.9.2010 and 1.10.–22.12.2010 from 6 to 39 samples per treatment in each period. The first and the last periods were shorter than the others. In this paper, the pore water concentrations are presented for the depths of 50 cm (sulfuric Bgic horizon) and 90 cm (sulfidic Cg horizon). Iron and Al concentrations have been presented in more detail in Virtanen et al. (2014a) but the comprehensive composition of the pore water, including Ca, Mg, K and Na, throughout the experimental period has not been published before.

In Ylistaro (62° 55' N, 22° 29' E), pipestems were collected from the depth of 60–190 cm and investigated with a scanning electron microscope (SEM). The elemental composition was determined using an ISIS energy dispersive X-ray EDX microanalysis system. The mineralogy of the pipestems was investigated by powder X-ray diffraction (XRD).

In Joensuu (62° 24' N, 30° 04' E), nine samples of the timothy vegetation growing in the area with no peat left were analysed for Ca, P, K, Cu, Zn, Mn, Fe, Cd, Cr and Ni. The plants were collected right before flowering, ashed at 550 °C, and the ash was dissolved in 0.2 M HCl. Soil samples were collected from the oxidized horizons (0–20 cm, 20–40 cm, 40–70 cm) and from the reduced subsoil (70–85 cm) and analysed for Zn bound to 1) easily soluble forms (acetic acid extraction), 2) sesquioxides (hydroxyl ammonium chloride), 3) organic matter (H₂O₂ followed by ammonium acetate) and 4) primary minerals (aqua regia) (Rauret et al., 1999).

2.2. Monitoring of emissions

In Joensuu, N₂O and CO₂ emissions from soil were monitored on the average at 2–3-wk intervals in the snow-free periods 18 May 2014–7 Dec 2014 and 13 May 2015–24 Aug 2015 along a transect at three locations with different depths of peat layer (15 cm at site 1, 30 cm at site 2 and 60 cm at site 3). The gas emissions were measured by closed chambers from which the samples were taken with a polypropylene syringe after 0, 20, and 30 min of closing the chambers, and injected into glass vials (3-ml Labco Exetainer® vials with double septa, Labco Ltd., UK). The gaseous composition was analysed by a gas

Table 1
Properties of the relevant soil profiles of the three experimental soils.

Helsinki, Sulfic Cryaquept	Clay	Silt	C	N _{tot}	S _{tot}	pH
	%	%	%	%	g kg ⁻¹	
Bgic 30–60 cm	61 ^b	34 ^b	1.9 ^c	0.26 ^d	4.2 ^c	3.8 ^c
Cg1 86–100 cm	58 ^b	40 ^b	2.8 ^c	0.40 ^d	20.8 ^c	6.5 ^c
Ylistaro, Sulfic Cryaquept						
Bgj1 52–70 cm	27 ^a	67 ^a	1.0 ^a	0.15	1.7	3.8 ^a
Bgj4 125–150 cm	26 ^a	65 ^a	1.3 ^a	0.21	3.1	3.4 ^a
Cg 210–215 cm	n.d.	n.d.	1.6	0.27	7.0	6.6
Joensuu, Typic Sulfaquept (timothy and Zn speciation)						
Ap 0–20 cm	10	43	6.3	0.07	5.7	4.6
Bg 20–40 cm	n.d.	n.d.	0.3	n.d.	n.d.	3.8
BCg 40–70 cm	n.d.	n.d.	0.3	0.08	5.9	3.9
Cg2 70–85 cm	n.d.	n.d.	n.d.	n.d.	n.d.	4.8
Joensuu, Typic Sulfisaprist (emissions)						
Op 0–25 cm	n.d.	n.d.	33.4	n.d.	n.d.	3.6
O1 25–50 cm	n.d.	n.d.	32.7	n.d.	n.d.	3.1
Cg 50–60 cm	19	69	0.5	0.10	7.1	4.5

^a Yli-Halla 1997.

^b Mokma et al., 2000.

^c Virtanen et al., 2014a.

^d Šimek et al., 2011.

chromatograph equipped with thermal conductivity (CO_2 , O_2 , N_2), flame ionization (CH_4 , C_2H_4) and electron capture (N_2O) detectors (Agilent Technologies 7890B GC custom, Santa Clara, CA, United States). Temporal drifts due to any changes in the sensitivity of detectors during long runs were accounted for by analyzing standard gas samples at 10-sample intervals (as detailed for N_2O by Penttilä et al., 2013; Appendix S1). The cumulative N_2O and CO_2 emissions were calculated for each chamber by a trapezoidal rule. For this purpose, a total of seven failed flux measurements for single chambers on different sites and measurement days were replaced by a median of other chambers on the same site. Groundwater level, temperature and volumetric water content in soil were monitored continuously at each site (Decagon Em50 loggers and 5TE sensors, Solinst 3001 Leveloggers and Barologger). Mineral N in soil was extracted with 2 M KCl solution from frozen soil samples and determined by colorimetry with a flow injection autoanalyser (LaChat QuikChem 8000) (see Šimek et al., 2011).

2.3. Other chemical analyses

The soils of all sites were analysed for total C and N (VarioMax CN), particle size distribution (pipette method), $\text{pH}(\text{H}_2\text{O}, 1:1)$ and total S for which the soil was digested with *aqua regia* (Helsinki, Joensuu) or 7 M HNO_3 (Ylistaro). The presence of sulfidic materials in the subsoils was confirmed by the decrease of pH upon aerobic incubation of 6 or 8 weeks. Properties of relevant soil horizons, partly compiled from earlier papers, are presented in Table 1.

2.4. Statistical analyses

Mean values and standard deviations were calculated for pore water composition and mineral element composition of timothy. The cumulative N_2O and CO_2 emissions from soil in Joensuu 2014 and 2015 were analysed statistically by a repeated measures ANOVA with the experimental year as a within-factor, and the measurement site (peat depth) as a between-factor, followed by multiple comparisons of means by a Sidak-adjusted LSD-test (IBM® SPSS® Statistics, Version 22). The relationships of gas emission rates and selected soil variables were investigated by calculating their Pearson correlation coefficients separately for each site.

3. Results and discussion

3.1. Pore water composition

In the lysimeter study performed with the Helsinki soil, low groundwater (LGW) level kept the Bgjc horizon oxidized and allowed slow oxidation of sulfidic materials in the Cg horizon at the bottom of the lysimeter during the summers (Virtanen et al., 2013). Instead, in the high groundwater (HGW) level treatment, the sulfidic Cg horizon was properly inundated, and the Bgjc horizon that had been oxidized in the field was also water-logged. By the end of the experiment, the pore water of the Bgjc horizon had a 1.5 units lower periodical mean pH in LGW than in HGW (Fig. 2). The low pH resulted in five times higher Ca and Mg concentrations in pore water of LGW than in the HGW treatment. However, Al concentration was high in this horizon in the LGW treatment during the entire experiment (Virtanen et al., 2014a). At the end of the experiment 57% of all pore water cations, expressed as $\text{mmol } (+) \text{ l}^{-1}$, in the LGW treatment of the Bgjc horizon were comprised of Al. After the initial increase, there was no long-term trend in the concentration of any cation in this horizon in the LGW treatment. In the inundated (HGW) Bgjc horizon, the base cations remained at a low and stable level throughout the experiment. Owing to higher pH, practically no dissolved Al was detected (Virtanen et al., 2014a). Instead, in the Bgjc horizon of the cropped HGW treatment, Fe gradually became the dominant cation in the pore water, 77% of all pore water cations being Fe at the end of the experiment. This Fe was released from the abundant

precipitates upon the development of reducing conditions (Virtanen et al., 2013, 2014a).

In the LGW subsoil containing sulfidic materials (Cg horizon), Ca and Mg concentrations increased 6–7 fold compared to the HGW subsoil by the end of the experiment. The subsoil pH of the LGW treatment decreased temporally in summers 2009 and 2010 below pH 5 (Virtanen et al., 2013) and the effective oxidation of sulfidic materials started in summer 2010 (Virtanen et al., 2014a, 2014b). This was indicated by a striking increase in the pore water Fe concentration in summer 2010, which can be seen also in Fig. 2d. Initially the exchange sites of subsoils were mainly occupied by base cations (Šimek et al., 2011). Therefore, the slow decrease in soil pH was evidently attributed to a cation exchange buffer as presented by Hartikainen and Yli-Halla (1986). Similarly, Ca and Mg increased also in pore water in the lysimeter study. However, in the Cg horizon the elements did not efficiently leach out due to the low K_{sat} of the subsoil (Virtanen et al., 2014b). They were abundantly found in the Bgjc horizon likely promoting transport of elements downwards by diffusion or out of the field to watercourses by convection. The pore water composition of the HGW subsoil remained stable throughout the experimental period. There, Na was by far the dominant cation (48% of pore water cations at the end), reflecting the recent marine origin of the soil. Because the oxidation of sulfidic material was prevented by waterlogging, much less mobilization of elements was observed in HWG than in LWG. The lysimeter experiment showed that in the Bgjc horizon water-logging effectively stops mobilization of Al (Virtanen et al., 2014b) and the same seems to happen with Ca and Mg as well. However, when previously oxidized horizons are water-logged again and reducing conditions develop, large amounts of Fe can be mobilized from the Fe hydroxide coatings of aggregates to the extent that it can become the dominant cation in the pore water. In this experiment the development of reducing conditions upon re-flooding was probably enhanced by the vigorous growth of plant roots into the Bgjc horizon and subsequent production of easily decomposable organic matter.

3.2. Composition of timothy

In the former peat excavation area of Joensuu the peat layer had been removed and the black schists had been oxidized rapidly. It is likely that the sulfidic materials originally extended to the top of the mineral soil, which now forms the plough layer. Owing to heavy liming of the Joensuu field, the timothy grew well. The topsoil had a pH of 4.6 but in the unlimed margins the topsoil pH was as low as 2.8–3.3. The concentrations of K, Cu, Mn and Fe in plants (Table 2) were the same as commonly found in Finnish timothy of the same growth stage while Ni and Cr concentrations were somewhat higher but not excessive. Higher Ni and Cr concentrations in timothy have also been observed earlier in AS soils (Palko, 1986). Plant P and Ca were rather low but did not suggest deficiency. No sign of excessive Cd concentration in plants was observed. The most prominent observation was a very low concentration of Zn in timothy. The concentration was close to deficiency level and only one third of what was the average Zn concentration in an earlier study (Kähäri and Nissinen, 1978) where 2000 timothy samples from Finland were analysed. This is surprising since low pH is commonly associated with high Zn concentrations in plants (Yli-Halla, 1993). Thus, this study did not support the assumption that plants growing in an AS soil would experience excessive supply of heavy metals.

3.3. Partitioning of heavy metals in the soil profile

Analysis of the Joensuu soil revealed that the concentration of the most soluble Zn fraction was small ($1.1\text{--}2.4 \text{ mg kg}^{-1}$, 3–9% of total Zn) in the three uppermost (0–70 cm) horizons but substantially higher (14.8 mg kg^{-1} , 23% of total Zn) at 70–85 cm (Fig. 3). This high proportion of Zn in the most labile fraction is similar to a polluted topsoil

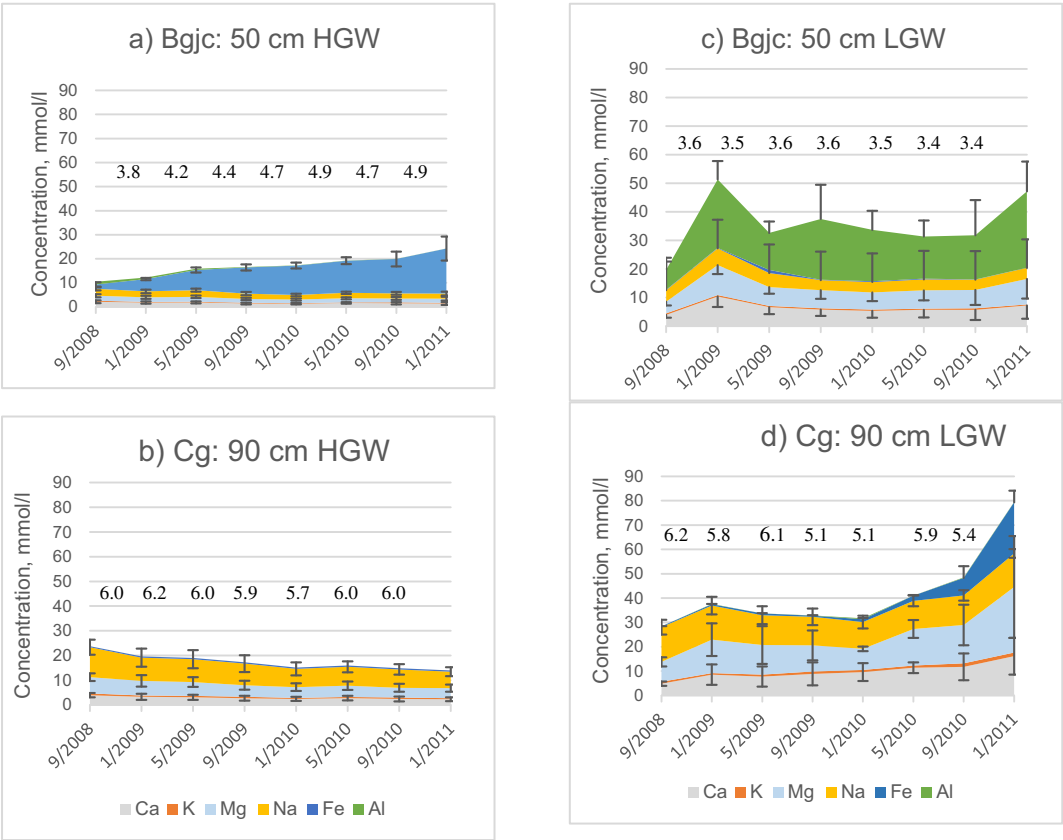


Fig. 2. Pore water composition in the oxidized horizon (50 cm) and in the reduced subsoil (90 cm) upon high and low groundwater (HW and LW, respectively) presented as averages with standard errors for 4-mo time intervals (the first and the last interval are shorter). The mean pH values for each interval are presented above the mean elemental concentrations. The Fe and Al concentrations in pore water are adapted from Virtanen et al. (2014a) and pH from the time series data of Virtanen et al. (2013).

Table 2

Mineral element composition of timothy grown in an AS soil in Joensuu and reference concentrations from other studies. The results of Palko (1986) come from AS soils while the others are from non-AS soils.

Element	Joensuu <i>n</i> = 9	Kähäri and Nissinen, 1978; Paasikallio, 1978 <i>n</i> = 2000	Palko, 1986 <i>n</i> = 45	Sillanpää and Jansson, 1991; Ylärinta and Sillanpää, 1984 Jansson et al., 1985 <i>n</i> = 9
Ca, g kg ⁻¹	1.60 ± 0.23	2.55	1.58	3.0
P, g kg ⁻¹	1.66 ± 0.23	2.91	1.93	2.85
K, g kg ⁻¹	20.2 ± 1.83	23.7	18.1	25
Zn, mg kg ⁻¹	11.9 ± 1.46	31.6	22.9	30
Cr, mg kg ⁻¹	0.86 ± 0.15	0.26	1.12	
Cd, mg kg ⁻¹	0.03 ± 0.01			0.06
Cu, mg kg ⁻¹	5.18 ± 0.77	4.24	3.76	6.2
Ni, mg kg ⁻¹	1.50 ± 0.51	0.36	1.83	
Fe, mg kg ⁻¹	38.9 ± 5.14	42.7	45.0	75
Mn, mg kg ⁻¹	66.1 ± 5.51	63.8	94.8	75

next to a metal smelter in Harjavalta, Finland (Kaasalainen and Yli-Halla, 2003) analysed with the same method. Also the oxalate-extractable fraction in the Joensuu soil was at least 2.5 times higher in the deepest horizon. Commonly acidity promotes solubility of Zn, but in this soil the highest concentrations of the easily soluble and oxalate extractable Zn occurred in the horizon which had the highest pH (4.8) while the lowest concentration occurred in the horizons with the lowest pH (3.8). This discrepancy suggests mobilization of Zn in the topsoil and subsequent leaching to subsoil and possibly further out of the soil profile. Depletion of Zn and subsequently low concentration of plant Zn

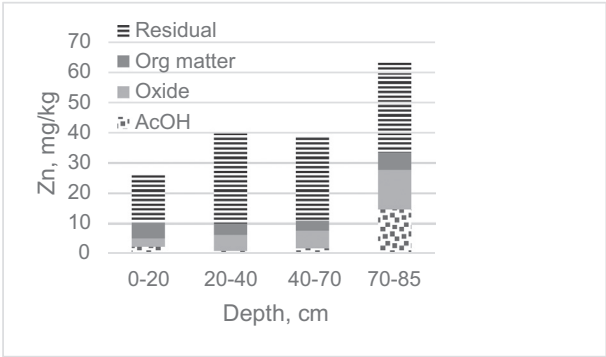


Fig. 3. Concentration of Zn in different fractions in a coarse silty Typic Sulfaquept in Joensuu. AcOH stands for acetic acid dissolving the most labile fraction.

has been promoted by even lower pH values than today that prevailed before the heavy liming of the field. The phenomenon of metal depletion was also documented by Claff et al. (2011) who observed decreased concentrations of Cu, Mn, Ni and Cr, but not Zn, in the oxidized AS soil horizons. On the basis of our results of plant and soil composition, the mobility of heavy metals in AS soils seems thus to be an environmental rather than an agricultural problem.

The assumed depletion of Zn in oxidizing AS soil was also supported by the results of the pore water composition in the lysimeter study in Helsinki (Fig. 4). In the most acidic Bgjc horizon (for pH, see Fig. 2) of the LW treatment the pore water Zn concentration was much higher than in the HW treatment during the entire monitoring. The maximum pore water concentration occurring in summer 2010 was seven times higher (average 2.5 mg l⁻¹) than in the HW treatment of the

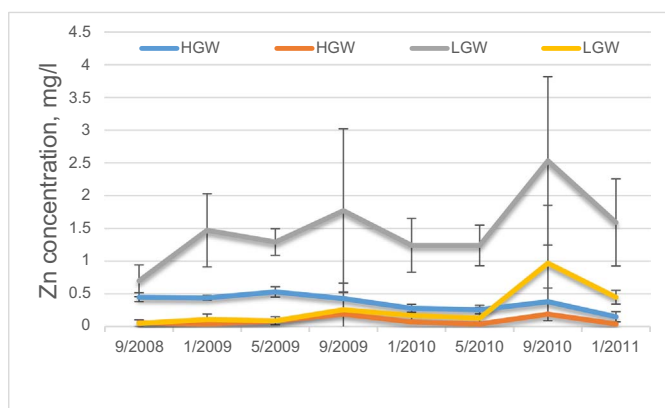


Fig. 4. Concentration of zinc in pore water in the oxidized horizon (50 cm) and in the reduced subsoil (90 cm) upon high and low groundwater (HGW and LGW, respectively) presented as averages with standard errors during 4-mo time intervals (the first and the last interval are shorter).

Table 3

The cumulative emissions of N₂O and CO₂ at the experimental sites of Joensuu 2014–2015.

Site (peat depth, cm)	Cumulative N ₂ O emission N g ha ⁻¹ yr ⁻¹			Cumulative CO ₂ emission C kg ha ⁻¹ yr ⁻¹		
	2014	2015	Mean	2014	2015	Mean
1 (15 cm)	647	1322	985 ^{ab}	5140	8389	6765 ^a
2 (30 cm)	890	4447	2669 ^b	3893	7130	5511 ^a
3 (60 cm)	469	261	365 ^a	4222	7051	5636 ^a
Mean	669 ^A	2010 ^B		4418 ^A	7523 ^B	
S.E.	247	1030	780	452	660	545

Repeated measures ANOVA (within-factor year) did not confirm significant interaction year × site for the cumulative emissions of N₂O or CO₂. For N₂O, the effects of year ($F_{(1, 9)} = 5.265$, $p = 0.047$) and site ($F_{(2, 9)} = 4.664$, $p = 0.041$) were significant, but the interaction year × site was non-significant ($F_{(2, 9)} = 3.782$, $p = 0.064$), with MSerror (Year) = 2,050,891 and MSerror = 2,436,655. For CO₂, only the effect of year was significant ($F_{(1, 9)} = 6.25463$, $p < 0.001$), while those of site ($F_{(2, 9)} = 3.213$, $p = 0.088$) and the interaction year × site ($F_{(2, 9)} = 1.234$, $p = 0.336$) were non-significant, with MSerror(Year) = 92,480 and MSerror = 1,186,788. Means denoted by the same letter do not differ at $p < 0.05$.

same horizon (average 0.37 mg l⁻¹). The lowest concentrations in the LGW treatment of this horizon occurred in winter when GW level also in this treatment was elevated. In the Cg horizon the Zn concentrations in the LGW were much lower than in the Bgic horizon but higher than in the HGW treatment. This difference may be attributed to the oxidation of sulfides in the LGW treatment because the elevation coincides with the striking increase of dissolved Fe in this horizon (see Fig. 2d). These results demonstrate the mobilization of Zn in the oxidized and highly acidic Bgic horizon and are in line with the higher concentrations of metals measured in water courses draining agricultural AS soils

Table 4

Daily groundwater level, and temperature and water content in topsoil at the experimental sites of Joensuu on the gas flux measurement days in 2014–2015.

Site (peat depth)	2014			2015		
	GWL	Temp	VWC	GWL	Temp	VWC
1 (15 cm)	43 (23...71)	8.8 (1.2...17.5)	0.55 (0.40...0.65)	35 (22...45)	10.6 (6.4...14.1)	0.53 (0.43...0.67)
2 (30 cm)	49 (31...74)	8.8 (1.1...17.5)	0.38 (0.21...0.50)	43 (32...52)	10.7 (5.2...14.6)	0.45 (0.29...0.72)
3 (60 cm)	82 (61... > 90)	9.5 (0.6...19.4)	0.39 (0.32...0.43)	73 (58...84)	12.5 (7.5...16.0)	0.43 (0.36...0.51)

GWL ground water level (cm below soil surface), Temp temperature at 7.5 cm soil depth (°C), VWC volumetric water content at 7.5 cm soil depth (m³ m⁻³). Flux measurement periods 18 May 2014–7 Dec 2014 (13 measurements) and 13 May 2015–24 Aug 2015 (6 measurements). Daily mean values (and the range of measured daily values) are shown.

(Åström and Corin, 2000). Sulfuric horizons can thus be important sources of dissolved Zn but the load to watercourses may be decreased by elevating the groundwater level.

3.4. Nitrogen stocks and emissions

In the Joensuu soil, the field section which still had 40 cm peat at the surface contained 270 kg mineral N ha⁻¹ in the upper 70 cm, 80% of it at 50–70 cm, which is an abundant amount compared to the average of 43 kg ha⁻¹ and range of 17–94 kg ha⁻¹ at 0–100 cm in 24 experiments in Finland, Sweden, Norway and Denmark (Lindén et al., 1992). Most of the stock consisted of NH₄⁺-N. It is commonly observed that the mineral N stock in the upper 100 cm of AS soils is not necessarily much higher than in other soils but deeper in the subsoil the AS soils have a large stock of NH₄⁺-N (Paasonen-Kivekäs and Yli-Halla, 2005; Šimek et al., 2011). In the depth of 0–100 cm of our Ylistaro soil, there was 135 kg mineral N ha⁻¹ in spring 2012 while at 160–190 cm alone, representing the transition horizon between the oxidized and reduced soil, the stock was as large as 250 kg ha⁻¹. However, like in other soils, only a small part of soil N stock in AS soils consists of mineral N. For example, in the Ylistaro soil the total N stock at 0–200 cm was about 50 Mg ha⁻¹, 1–2% being in mineral form. Paasonen-Kivekäs and Yli-Halla (2005) calculated that in spite of apparently large amounts of NH₄⁺-N, < 2% of cation exchange sites were occupied by this ion in the horizon where the concentration was the highest.

The cumulative emissions of N₂O during snow-free periods were only 0.5–0.9 N kg ha⁻¹ in 2014 and 0.3–4.4 N kg ha⁻¹ in 2015 (Table 3). Least N₂O was emitted from site 3 (60-cm peat), and most from site 2 (30-cm peat), whereas the CO₂ emissions were unaffected by the depth of peat layer. Despite the significant year-to-year variation, the magnitude of N₂O emissions was very modest for a peat soil. Only at site 2 (30-cm peat), can the magnitude of N₂O emissions be considered typical of cultivated boreal peat soils. The emissions reported in Table 3 cover only part of the year, but even if calculated on a 365-day basis, the average annual emission would correspond to only 3.2 kg N ha⁻¹. This is only about 1/10 of what has been measured in a coastal AS soil in W Finland (Uusi-Kämpä et al., 2012).

The reasons behind the diverse N₂O emissions at the different sites remain to be determined. Even if the groundwater levels were rather high at all sites of the field, they did not generally rise to the O horizon but stayed in the upper parts of B horizon. The relatively low ground water levels at site 3 (60-cm peat layer) may have contributed to the lower emissions found there (Table 4). Further, correlation analysis was used to study if the limiting factors for the emissions differed between the sites during the experimental period. The correlations between N₂O emission rates and the monitored soil variables (temperature, moisture content, groundwater level) were weak and mostly non-significant at all sites (Table 5). At site 1, the N₂O emissions increased significantly with increasing topsoil moisture content, and the high N₂O emissions at the time of frost melt probably contributed to the significant negative correlation between N₂O emissions and soil temperature at 22.5 cm

Table 5
Correlation between the N₂O and CO₂ emission rates and selected soil variables in Joensuu in 2014–2015.

Variable	Site 1		Site 2		Site 3	
	N ₂ O emission rate	CO ₂ emission rate	N ₂ O emission rate	CO ₂ emission rate	N ₂ O emission rate	CO ₂ emission rate
N ₂ O emission rate	1	– 0.067	1	0.001	1	0.226*
CO ₂ emission rate	– 0.067	1	0.001	1	0.226*	1
Depth of GWL	– 0.208	0.665**	– 0.141	0.390**	0.094	0.336**
VWC at 7.5 cm	0.237*	– 0.442**	0.008	– 0.324**	0.126	– 0.132
VWC at 22.5 cm	– 0.222	– 0.567**	– 0.083	– 0.355**	0.015	– 0.463**
VWC at 45 cm	– 0.078	– 0.134	– 0.083	– 0.502**	0.216	– 0.287*
Temp at 7.5 cm	– 0.126	0.607**	0.077	0.722**	0.008	0.683**
Temp at 22.5 cm	– 0.345**	0.503*	0.046	0.560**	– 0.172	0.457**
Temp at 45 cm	– 0.078	– 0.134	– 0.083	– 0.502**	0.216	– 0.287*

Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. $n = 72...76$ (significant correlations shown in bold).

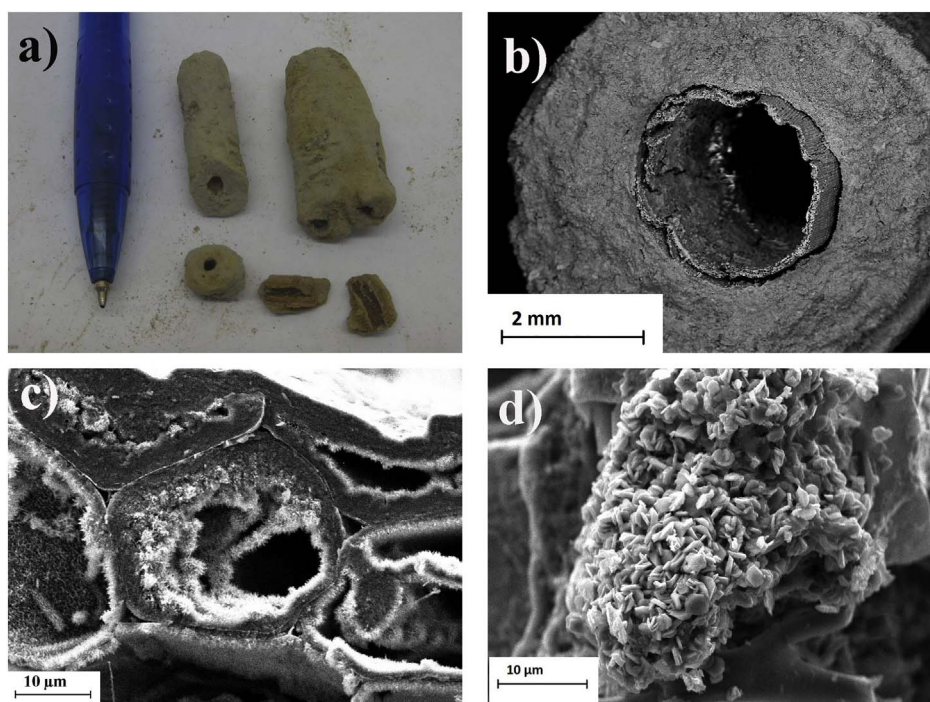


Fig. 5. Pipestems of the Ylistaro fine silt soil: a) pipestems by the naked eye, b) a pipestem with the remaining cortex cell walls at 60–80 cm, c) cortex cells half-filled with schwertmannite at 170–190 cm, d) jarosite in the pipestem matrix at 130–150 cm.

depth. In contrast, the CO₂ emission rates increased with increasing topsoil temperature, decreasing groundwater level and decreasing soil moisture content at all sites, as expected (Table 5). The N₂O and CO₂ emission rates showed a significant positive correlation only at site 3. Assuming the N₂O production in the soil was mainly driven by microbial processes, the decreased biological activity by low pH and high metal content in soil may partly explain the small emissions. However, as the role of pH is not limited to the overall biological activity but may also affect differently the microbial species and the functioning of denitrifying enzymes, and as the magnitude of N₂O production by chemodenitrification in AS soils is largely unknown, the mechanisms and limiting factors of N₂O production and emission remain to be investigated in more detail in the follow-up studies.

3.5. Composition of pipestems

In the Ylistaro silty clay loam soil pipestems were found between 60 and 190 cm (Fig. 5a). Around the central hole of many pipestems the cortex of the former root was still visible (Fig. 5b). In the pipestem matrix, consisting of silicate particles cemented together, there was 6.8–13.8% Fe and 19.9–26.1% of Si. There was thus accumulation of Fe oxides in this oxidized zone surrounded by the reduced bulk soil where the Fe content was only 2.8–3.3%. It has also been shown that the

pipestems taken from the Bgjc horizon of the Helsinki clay soil contained as much as 22 times more oxalate extractable Fe than the surrounding bulk soil (Yli-Halla et al., 2008). In the interior of the pipestems, a new solid phase was precipitated within the former cortex cells (Fig. 5c). According to the EDS analysis the precipitates in the cell interiors contained as much as 43–53% of Fe and only 1–4% of Si, indicating that this solid phase did not consist of silicate minerals. The XRD analysis and SEM image (Fig. 5d) showed that jarosite $KFe_3(OH)_6(SO_4)_2$ was clearly present in the soil but the EDS data and the mineral morphology suggested that the cortex cells were predominantly filled with schwertmannite, $Fe_8O_8(OH)_6(SO_4)_nH_2O$.

4. Conclusions

Upon oxidation of sulfidic materials and subsequent acidification, Ca and Mg seem to be the major base cations to be mobilized. Water-logging prevented or stopped mobilization of those elements. Timothy growing in an AS soil showed only small deviations from the average composition measured in Finland with the exception of Zn that was at the deficiency level. This may be attributed to the stripping of Zn from the root zone. Large amounts of Zn had probably been dissolved in the sulfuric horizons and subsequently leached. This observation was supported by much lower Zn concentration in the oxidized topsoil as

compared to water-logged Cg horizon. Pipestems that had developed in previous root channels indicated accumulation of Fe and formation of a new solid phase probably mostly in the form of schwertmannite even though jarosite was also detected. The excess of Fe and sulfate, created by the rapid oxidation of sulfidic minerals, may be conducive to the formation of schwertmannite in our silty soil probably of low K supply. Several elements except Fe are thus mobilized as a response of the oxidation of sulfidic materials but water-logging prevents the process. The amount of N lost as N₂O from the black schist derived AS soil which had a peaty topsoil was relatively small, the emissions being at a low level more typical of mineral rather than organic soils. Large mineral N stocks of an AS soil thus appear not necessarily to contribute to abundant gaseous N₂O emissions. The low correlation of N₂O emissions with CO₂ emission or measured soil variables emphasises the importance to identify and study the mechanisms and limiting factors of N₂O production and emission in future studies.

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